

The Incentives of Scientific Experts: Evidence from the History of Public Health: Appendix

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A Cholera Mortality

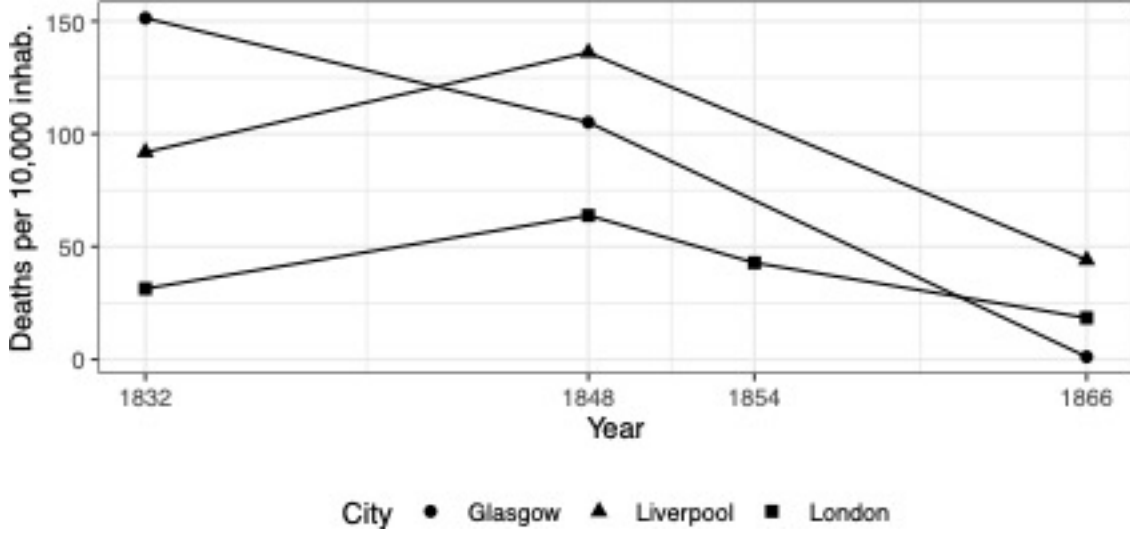


Figure A1: Mortality rates for selected cities during nineteenth-century cholera pandemics. Data for Glasgow comes from Underwood (1948) and University of Portsmouth (2017); for Liverpool from Underwood (1948) and Battersby (2017); and for London from Underwood (1948), Royal Society of Arts (1878), and Haughton (1867). In the context of British mortality, cholera stood out as a threat in urban centers, though its overall impact on national mortality rates was more muted. No epidemic had caused a major spike in deaths for several generations (the most recent was an outbreak of typhus in 1740) (Wrigley and Schofield 1989).

B A Simple Model of Experts’ Dual Incentives

Suppose that a scientific community is interested in knowing the the state of the world $\omega \in \{-1, 1\}$. They have a common prior $\pi = P(\omega = 1) = 1 - P(\omega = -1)$.¹ An expert searches for new evidence, which takes the form of a signal $s \in \{-1, 0, 1\}$. With probability p , the expert uncovers a signal $s \in \{-1, 1\}$. The signal s and the associated probability $P(\omega|s)$ (for an informative signal $s \in \{-1, 1\}$) satisfy the monotone likelihood ratio property such that s gives useful, but not infallible, information about ω : for $\omega \in \{i, j\}$, $1 > P(s = i|\omega = i) > P(s = j|\omega = i) > 0$, so it is possible to observe a “mistaken” signal that does not match ω . With probability $1 - p$, the expert fails to find any useful evidence, returning a signal $s = 0$.

After searching, the expert transmits a message $m \in \{-1, 0, 1\}$ to the scientific community. An expert who finds a signal $s \in \{-1, 1\}$ has the opportunity to reveal it or can reveal nothing (i.e., $m = 0$, sending an “empty message”). However, the expert cannot falsify evidence he or she did not find; if the expert found no evidence ($s = 0$), he or she can only send an empty message, and an expert who received a signal $s = -1$ cannot send a message

¹This prior can be thought of as representing either the community’s certainty or, if it represents the average view, the degree of consensus.

of $s = 1$ (or *vice versa*).² Experts cannot commit *ex ante* to any strategy to reveal or hide their signals. After observing the contents of the expert’s message, the community updates its common prior to a new posterior μ (the community’s process for updating its belief is described below).

There are two types of experts: “unconflicted experts” (“he”) and “conflicted experts” (“she”). An unconflicted expert’s payoff $\hat{U}$ is solely a function of whether he sends a non-empty message, which represents, for instance, rewards to career advancement or social status from publishing a scientific finding:

$$\hat{U} = \begin{cases} 0 & m = 0 \\ \alpha & m \in \{-1, 1\} \end{cases} \quad (\text{A1})$$

A conflicted expert receives the same payoff as an unconflicted expert for sending a non-empty message. However, she also gets a payoff when the community’s belief is below a particular threshold θ (at which point, for instance, a certain policy preference that she favors is adopted):

$$\tilde{U} = \hat{U} + \begin{cases} \beta & \mu \leq \theta \\ 0 & \mu > \theta \end{cases} \quad (\text{A2})$$

Any expert’s type is common knowledge, as are the values of the parameters α and β .³ However, the community may have imperfect information about whether an expert who sent an empty message actually has no information to transmit.

The solution concept is a perfect Bayesian equilibrium in which an expert and a policymaker each update their beliefs in a manner that is sequentially rational and consistent on the equilibrium path. I focus on identifying the conditions under which a “truth-telling equilibrium” exists, where any expert, including a conflicted one who receives an unfavorable signal $s = 1$, reveals a message that matches his or her signal.

An unconflicted expert who sends an empty message gets a payoff of $\hat{U} = 0$ vs. a payoff of $\hat{U} = \alpha$ when he sends an informative message. Thus, he has a dominant strategy of sending an informative message whenever he can. Additionally, a conflicted expert has a dominant strategy of always conveying a message containing a “convenient” signal ($m = -1$) whenever possible, since doing so yields a minimum payoff of α without any chance of negative consequences for her policy preference.

Under what circumstances will a conflicted expert send the message $m = 1$? There are some situations, based on the values of the primitives α , β , θ , and π , in which even a conflicted

²Practically speaking, experts can present information they know to be false, but the professional penalties for doing so tend to be very high.

³Making type common knowledge diverges from a classic family of signalling models, such as Spence (1978), in which experts’ type is hidden, but in the context of experts with potential conflicts of interest, the conflict is often obvious. For instance, experts employed by a pharmaceutical company seeking FDA approval for a new product are known to be conflicted in that they benefit when the product is approved.

expert's strategy is to reveal a message consistent with $s = 1$. If $\pi > \theta$, the community's prior belief is already above the threshold. An expert who suppresses a signal $s = 1$ cannot move the belief below the threshold. The expert thus reveals $m = s = 1$ for a payout of $\tilde{U} = \alpha$ (vs. $\tilde{U} = 0$ if she does not reveal). If π is sufficiently far below θ , the expert can safely reveal $m = s = 1$ without causing μ to rise above the threshold.

In a truth-telling equilibrium, if the community observes $m = 1$, they update their common prior in a typical Bayesian fashion, since signals cannot be falsified:⁴

$$\mu = P(\omega = 1|s = 1) = \frac{P(s = 1|\omega = 1)\pi}{P(s = 1|\omega = 1)\pi + P(s = 1|\omega = -1)(1 - \pi)} \quad (\text{A3})$$

Setting Equation A3 equal to θ and solving for π yields

$$\pi = \frac{\theta P(s = -1|\omega = -1)}{(1 - \theta)P(s = -1|\omega = 1) + \theta P(s = -1|\omega = -1)} \quad (\text{A4})$$

In the truth-telling equilibrium, the expert can reveal an “inconvenient” signal without risking policy consequences. This holds in two situations. The first is for all π greater than θ . in which case she will reveal $m = s = 1$ for a payoff of α (vs. 0 if she did not reveal). The second is for all π lower than a value π^* , which is the highest prior at which revealing an inconvenient signal will still result in a posterior belief below θ . She will reveal $m = s = 1$ for a payoff of $\alpha + \beta$ (vs. β if she did not reveal).

On the other hand, if $\pi^* < \pi < \theta$, a conflicted expert who has a signal $s = 1$ may be forced to choose between her two sources of utility. Revealing her signal yields $\tilde{U} = \alpha$, since $\mu > \theta$. Can she improve her outcome? Not if $\alpha > \beta$; in this case, she is better off revealing the signal to get credit for the discovery.

To summarize, at least one of the following conditions must hold for a conflicted expert to always share a signal $s = 1$:

1. $\pi > \theta$ or $\pi < \pi^*$: either the audience must put sufficient weight on the possibility that $\omega = 1$ so that the expert's preferred solution is out of reach, or the audience must put sufficiently little weight on the possibility that $\omega = 1$ that the expert can safely reveal inconvenient information without risking policy consequences. Figure 1 in the main paper provides a visual that captures this intuition.
2. $\alpha > \beta$: the expert gets high enough credit for revealing a signal than it outweighs the benefits of her preferred policy.

⁴Analogously, if $m = -1$,

$$\mu = P(\omega = 1|s = -1) = \frac{P(s = -1|\omega = 1)\pi}{P(s = -1|\omega = 1)\pi + P(s = -1|\omega = -1)(1 - \pi)}$$

C Medical Vocabulary in the Nineteenth Century

Medical vocabulary in the nineteenth century was non-standardized; definitions of even fundamental terms varied over time and, sometimes, between authors active at the same time. Of key importance to our setting is what scientists meant when they used the terms “contagious” and “infectious” – two of the most common words used to describe disease transmission. (Other common words, such as “communication” and “transmission,” are clearly comprehensible to a layperson and avoid complicated ambiguity.)

Some authors used “contagious” and “infectious” interchangeably to denote a disease that is spread by its carriers. Others drew a distinction between “contagion,” which usually (but not always) meant transmission via direct contact, and the broader category of “infection” which usually (but not always) encapsulated indirect contact (such as via objects that had been touched by a patient, or waterborne disease transmission by way of a patient’s “effluvia”). For instance, an early government report in India in 1824 could state, “It may consequently be inferred, that the disease has either been propagated by infection or contagion; or, that its progress is owing to circumstances beyond our knowledge, thus ranking cholera amongst many other epidemics, the cause of whose origin and progress are equally unintelligible and unknown,”⁵ differentiating between the two, while a different author could write that “I shall endeavour to prove that malignant Cholera is infectious, according to the important principle laid down by Sydenham, ‘that all epidemic diseases of a malignant nature are capable of propagating themselves by contagion,’ ”⁶ using “contagious” and “infectious” more or less interchangeably. This highlights the importance of coding outcomes based on the author’s view of the key issue at stake: whether cholera was transmitted (directly or indirectly) by the movement of people.

Some other common words used to discuss cholera include “epidemic” – which in its nineteenth-century context usually referred to a disease that affected people in a specific geographic area, without necessarily indicating anything about its transmission mechanism (for instance, a “miasmatic” disease could be “epidemic” in a particular area) – or “pestilential,” by analogy to the plague, which was generally considered to be contagious. “Zymotic” is a term arising from early forays into microbiology to refer to potentially disease-bearing microscopic organisms, sometimes in the context of a theory that linked their multiplication to a fermentation process. Below, I give definition for these terms from two medical dictionaries, an 1839 edition of Robert Hooper’s *Lexicon Medicum* and an 1887 edition of Richard D. Hoblyn’s *A Dictionary of Terms Used in Medicine and the Collateral Sciences*, to give a greater sense of how medical vocabulary was used and evolved.

Contagion

Hooper 1839: CONTAGION. (Contagio , onis, f. ; from con and tango, to touch.) The term contagion has been used in several acceptations. 1. It has been employed to signify the

⁵Scot, William. (1824) “Report on the epidemic cholera as it has appeared in the territories subject to the presidency of Fort St. George,” xlvii

⁶Sadler, Michael Thomas. (1848) “A few thoughts on cholera: in reference to its origin, the nature of the exciting cause, and the principle of treatment,” 4.

communication of a disease by personal contact with the sick. 2. It has been employed to signify the communication of a disease, either by personal contact with the sick, or by an effluvium evolved from the body of the sick. 3. It has been employed as generic term, embracing all atmospheric and morbid poisons ; all the effluvia, miasmata, infections, and poisons that cause fevers, of whatever kind ; and those poisons which uniformly excite the diseases which give birth to them, as the venereal disease, the itch, tinea capitis, &c. The second of these senses is the one which it is now generally employed. Attempts have been made to distinguish between contagion and infection, the former being restricted to the communication of disease by direct contact, and the latter to that by effluvia arising from the body of the sick, and communicated through the medium of the atmosphere; but this distinction is now properly discarded by the majority practical writers, and the two words are considered as synonymous

Let it be understood, then, that contagion, [like] infection means the communication of a disease by personal contact with the sick, or by means of an effluvium arising from the body of the sick. . . (452)

Hoblyn 1887: CONTAGION (contagiim, for contagio, a touching, from contingere, to touch, take hold of). This term, and Infection, generally denote the transmission of a poisonous principle. When the transmission is effected by a material substance, and is brought about by actual contact, the term contagion (immediate contagion) is employed; but when transmission is effected through the agency of the winds, and at a distance, the mode of communication is called infection (mediate contagion). In other words, when the poisonous principle is volatile and communicable through the medium of the atmosphere, it is infectious; when this diffusibility is absent, it is contagious. (168)

Epidemic

Hooper 1839: EPIDEMIC. (Epidemicus, from [epi], upon, and [demos], the people.) Applied to a disease which attacks a multitude of persons at the same time and in the same place. Epidemic diseases may arise from contagion or from some atmospheric cause, or from both combined, which last is probably the case for most instances. (591)

Hoblyn 1887: EPIDEMIC ([epidemios], prevalent among a people). An epithet for a popular, prevailing, but not native disease, arising from a general and temporary cause, as excessive heat. (See *Endemic*.) The phrases “Epidemic constitution,” “Epidemic influences” belong to the earlier ages of physic, and are suggestive of mystical notions. “It is the disease that constitutes the epidemic, and not the epidemic the disease. The evil always remains the same, the number of those affected being alone increased.” (244)

Infection

Hooper 1839: See Contagion. (763)

Hoblyn 1887: INFECTION (inficere, to stain). A general term for the contamination of the atmosphere by malaria, by matter of contagion, by effluvia arising from putrid animal and vegetable substances, &c. See *Contagion*. (365)

Pestilential/Pestis

Hooper 1839: PESTILENTIAL. (*Pestilentialis*; from *pestis*, the plague.) A disease which is epidemic and malignant, partaking of the nature of a plague, is called a pestilential disease. (1011)

From the entry for PESTIS: The plague is by most writers considered as the consequence of a pestilential contagion, which is propagated from one person to another almost solely by *contact* with a diseased person, or with fomites embued with the specific poison. (1012)

Hoblyn 1887: PESTIS: This term is seldom used by good authors to signify an infectious disorder. It generally denotes a noxious atmosphere, destruction, curse, &c. (536)

Zymotic

Hooper 1839: No entry.

Hoblyn 1887: ZYMOTIC DISEASES ([zymotikos], causing to ferment, from [zyme], leaven). A term suggestive of a fermentation in the blood, occasioned by the introduction into the system of a specific or peculiar virus. The term is used synonymously with “acute specific diseases.” (806)

D Cholera in Parliament

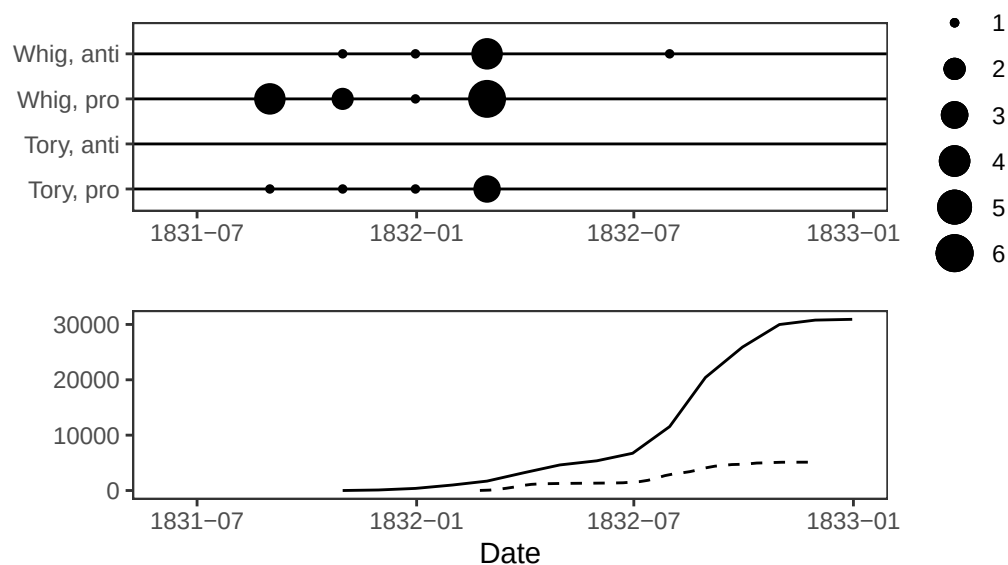


Figure A2: The top panel shows the number of pro- and anti-quarantine speeches made in Parliament by members of the Whig party (including Liberal and Radical members) versus the Conservative (Tory) party by date, contemporaneously plotted against all-Britain (solid) and London (dotted) cumulative cholera deaths in the bottom panel. Parliament had no further meetings after August in 1832.

E Raw Data

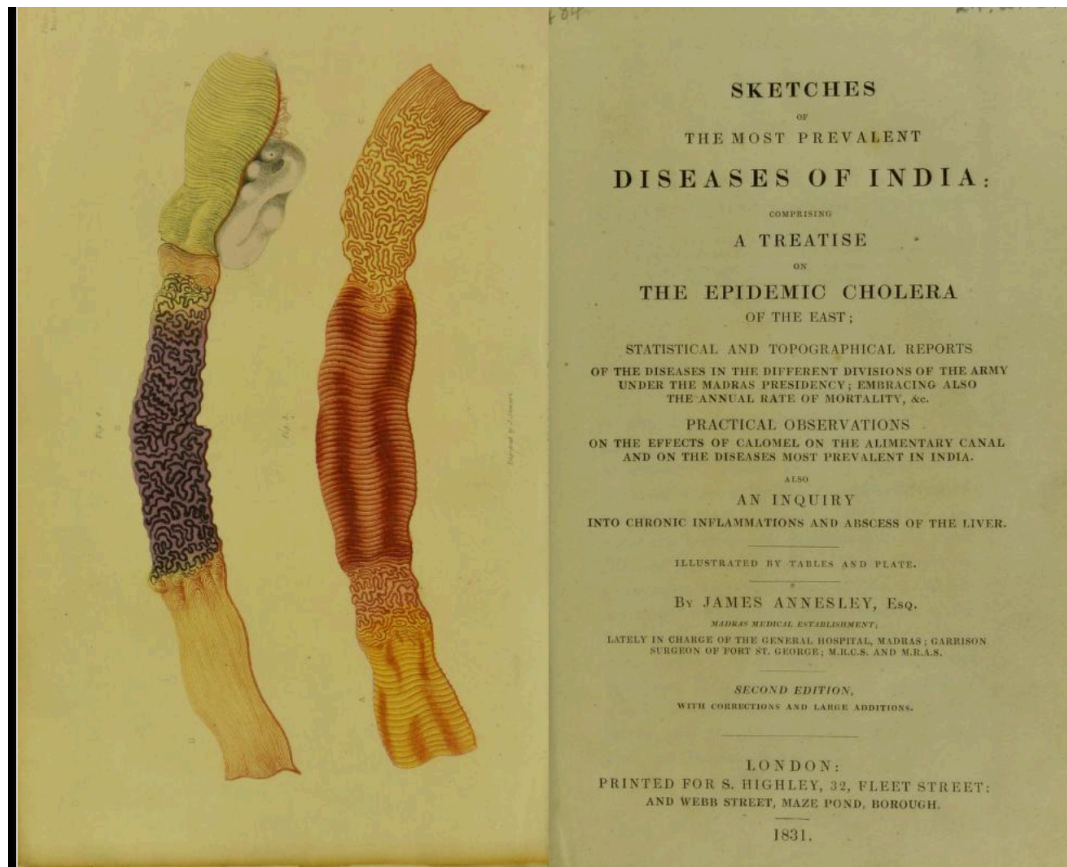


Figure A3: The frontispiece of MHL document b21306333 by James Annesley, a surgeon working in British India, in 1831.

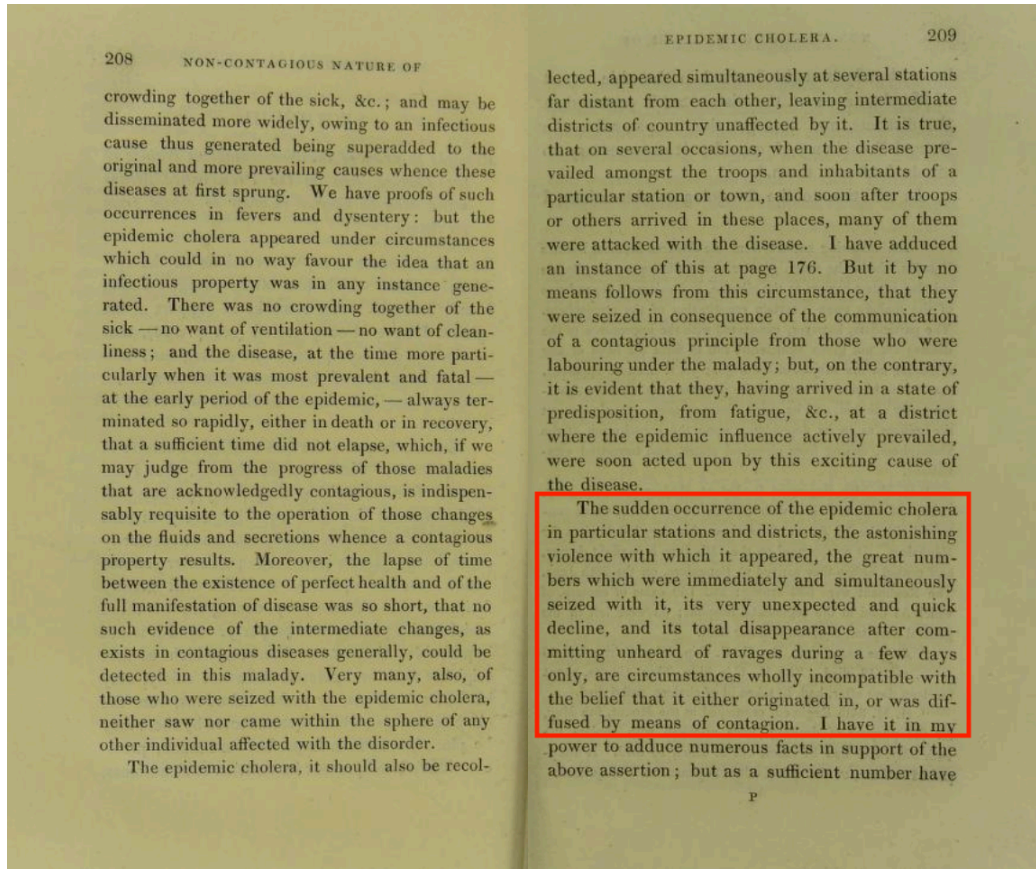


Figure A4: P. 209 from Annesley’s work, in which he writes that ‘The sudden occurrence of the epidemic cholera in particular stations and districts, the astonishing violence with which it appeared, the great numbers which were immediately and simultaneously seized with it, its very unexpected and quick decline, and its total disappearance after committing unheard of ravages during a few days only, are circumstances wholly incompatible with the belief that it either originated in, or was diffused by means of contagion.’ The document is accordingly coded as anti-contagionist.

F Sample Construction

Figure A5 shows the step-by-step process of cleaning and constructing the final sample of corpus documents. As of the time when data was scraped from the Medical Heritage Library online archive, there were over 127,000 documents with dates of publication in the nineteenth century (although the archive is continuously growing in size). Of these, just over 80,000 are in English. Each document includes one or more contributor-supplied metadata tags. 1,049 documents – about 1.3% of the entire sample – included “cholera” as such a metadata tag. I supplement these with an additional 113 documents that, while not containing a relevant metadata tag, frequently mentioned the word “cholera” in the actual text, which I define as greater than or equal to a frequency of 3.789 mentions per thousand words (the 10th decile frequency of the metadata-tagged part of the corpus). Together, these generated 1,162 documents. Eliminating English-language documents from outside of Britain

(mostly from the United States) and documents authored by non-experts or those not attributable to individual authors yielded a final sample of 418 documents.

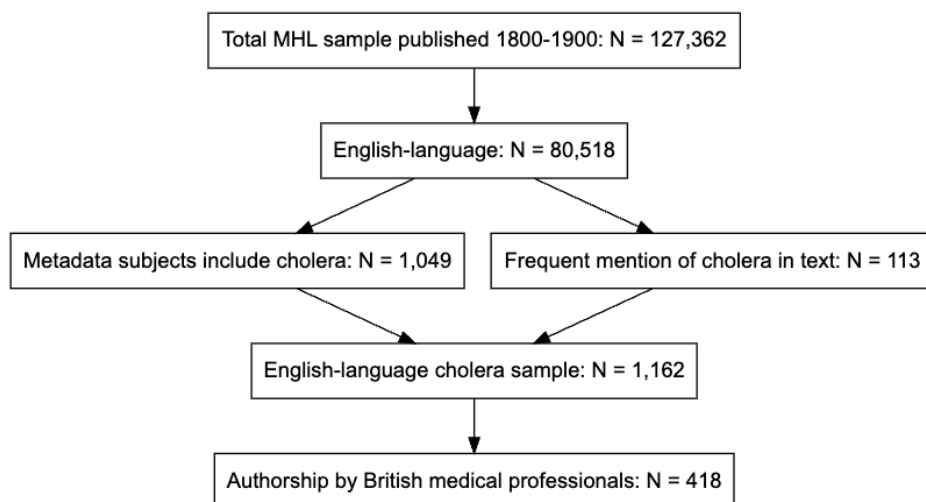


Figure A5: Flowchart showing construction of final corpus from Medical Heritage Library archival materials.

G Summary Statistics

Tables A1, A2, and A3 give summary statistics for trade-linked vs. non-trade-linked document-author observation pairs used in the main specifications. Consistent with the specifications used to generate the main results, the differences-in-means are weighted by the inverse of the number of authors to which a document is attributed.

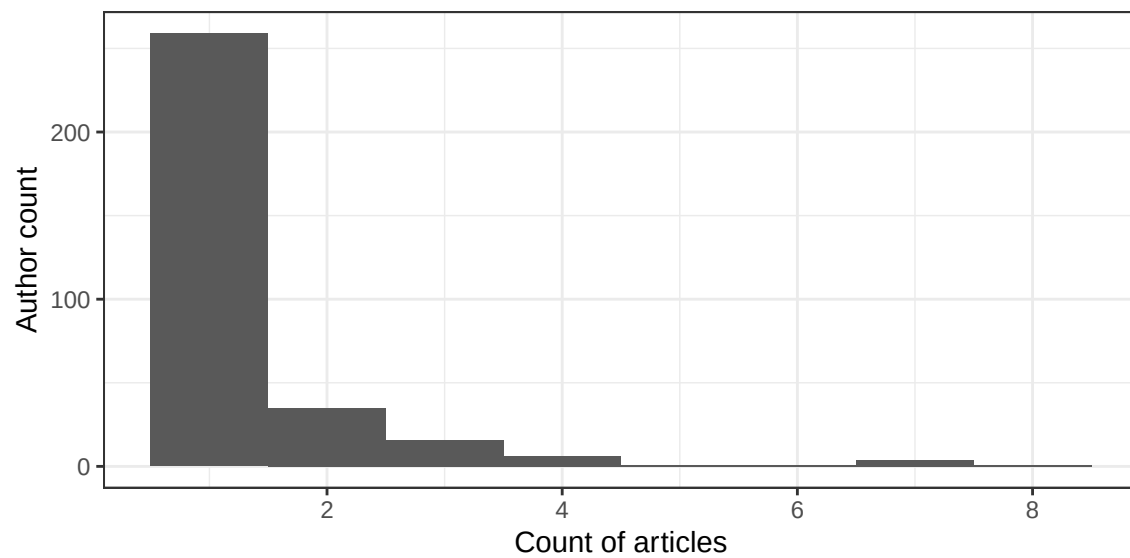


Figure A6: Count of authors by number of articles contributed to final sample.

Table A1: Summary statistics for trade-linked and non-trade-linked document-expert pairs, full sample.

	Trade=1 (N=103)					Trade=0 (N=345)					P-Value
	Mean	Med.	SD	Min.	Max.	Mean	Med.	SD	Min.	Max.	
FRCP	0.043	0	0.203	0	1	0.093	0	0.291	0	1	0.109
Prof. orgs.	0.702	0	0.971	0	4	0.976	1	1.158	0	7	0.035
Public audience	0.078	0	0.269	0	1	0.087	0	0.282	0	1	0.833
Age	47.363	47	12.824	24	86	47.76	45.5	14.232	20	100	0.948

Table A2: Summary statistics for trade-linked and non-trade-linked document-expert pairs, pre-1854 sample.

	Trade=1 (N=26)					Trade=0 (N=153)					P-Value
	Mean	Med.	SD	Min.	Max.	Mean	Med.	SD	Min.	Max.	
FRCP	0	0	0	0	0	0.069	0	0.254	0	1	0.238
Prof. orgs.	0.7	1	0.733	0	2	0.814	1	0.874	0	4	0.597
Public audience	0.154	0	0.368	0	1	0.124	0	0.331	0	1	0.632
Age	43.222	41	15.299	24	82	44.548	41	13.819	20	100	0.679

Table A3: Summary statistics for trade-linked and non-trade-linked document-expert pairs, post-1854 (and inclusive of 1854) sample.

	Trade=1 (N=77)					Trade=0 (N=192)					P-Value
	Mean	Med.	SD	Min.	Max.	Mean	Med.	SD	Min.	Max.	
FRCP	0.054	0	0.228	0	1	0.112	0	0.317	0	1	0.136
Prof. orgs.	0.703	0	1.03	0	4	1.102	1	1.326	0	7	0.018
Public audience	0.052	0	0.223	0	1	0.057	0	0.233	0	1	0.874
Age	48.384	49	12.04	29	86	50.209	49	14.092	25	85	0.484

H Inter-Coder Reliability of Outcome Variable

Documents that mentioned cholera were independently re-coded by a research assistant to ensure that views could be extracted from the documents with reasonable reliability (two observations were coded later and were omitted from the recoding). Table A4 shows the resultant confusion matrix. In cases of disagreement, the main coder (the author) re-examined each discrepant observation without reference to either the original coding or independent coding and issued a new revised classification. Overall, at the end of this process, the two coders agree 87.68% of the time. In cases in which neither coder categorized an entry as “neutral,” the two are in agreement 92.93% of the time.

The Cohen’s kappa, which accounts for chance agreement between coders, is 0.795, a level characterized by Landis and Koch (1977) as falling just on the border between “substantial” and “almost perfect” agreement. Table A5 shows that the main results change little if the second coder’s outcomes are used.

Table A4: Confusion matrix showing original coding (left) and second independent coding (top).

	Anti	Neutral	Pro
Anti	97	4	5
Neutral	6	43	4
Pro	15	9	166

I Robustness Checks

Table A5 shows robustness checks for the results shown in Table 2 of the main text. Column 1 swaps the category of interest in the outcome to contagionist, pooling anti-contagionist, neutral/unsure, and does-not-mention into the reference category, while Column 2 subsets the sample to only documents that are coded as clearly contagionist or anti-contagionist, omitting the neutral/unsure and does-not-mention categories. The smaller and noisier point effect in Column 1 is unsurprising, given that many articles that do not mention cholera do so precisely because the pro-contagionist consensus was established. Column 3 presents a logistic regression. Column 4 omits the author-per-document weights. Column 5 uses the second independent coder’s outcome. Column 6 uses period fixed effects instead of continuous date controls. (Year fixed effects are impractical due to sparse sampling in some particular years; periods are based on epidemic waves as defined in Figure 3 of the main text, and documents that are published between epidemics are assigned to the previous wave of disease.)

Table A5: Robustness checks for Table 2.

	<i>Dependent variable:</i>					
	Pro-contagionist stance	Subsample	Anti-contagionist stance			Period FE
	Swap outcome		Logistic	Unweighted	Alternate coder	
	(1)		(3)	(4)	(5)	(6)
Trade link	−0.109 (0.076)	0.200 (0.079)	1.023 (0.316)	0.187 (0.056)	0.156 (0.059)	0.180 (0.059)
Date	−0.537 (0.253)	−0.014 (0.281)	−0.772 (1.270)	−0.216 (0.202)	−0.199 (0.223)	
Date ²	0.0001 (0.0001)	0.00000 (0.0001)	0.0002 (0.0003)	0.0001 (0.0001)	0.0001 (0.0001)	
FRCP	−0.026 (0.122)	−0.120 (0.117)	−0.729 (0.535)	−0.110 (0.069)	−0.167 (0.072)	−0.108 (0.070)
Memberships	0.012 (0.030)	0.018 (0.039)	0.134 (0.166)	0.023 (0.028)	0.025 (0.023)	0.023 (0.030)
Public audience	0.004 (0.094)	0.030 (0.121)	0.284 (0.483)	0.057 (0.099)	0.066 (0.097)	0.061 (0.102)
Controls	Core	Core	Core	Core	Core	Core
Observations	426	299	426	426	424	426
Adjusted R ²	0.014	0.047		0.052	0.045	0.058
Log Likelihood			−222.932			
Akaike Inf. Crit.			459.864			

J Use of Bacteriological Vocabulary

Figure A7 shows the share, by year, of documents using at least one word stem that references bacteriological concepts (“animalcul-”, “vibrio-”, “bacil-”, or “bact-”). The dashed line denotes 1854 (the year Snow published proof of the waterborne nature of cholera and Pacini identified the relevant bacterium under a microscope). The size of each point is scaled to reflect the number of articles published by year. While some early research theorized about the role of invisibly small life forms in the spread of disease (typically termed “animalcules”), these theories could not be evaluated until advances in microscopy allowed for direct observation. The work of Snow and those who followed in his footsteps helped to elucidate the causal (vs. correlative) role of microscopic organisms in the spread of disease.

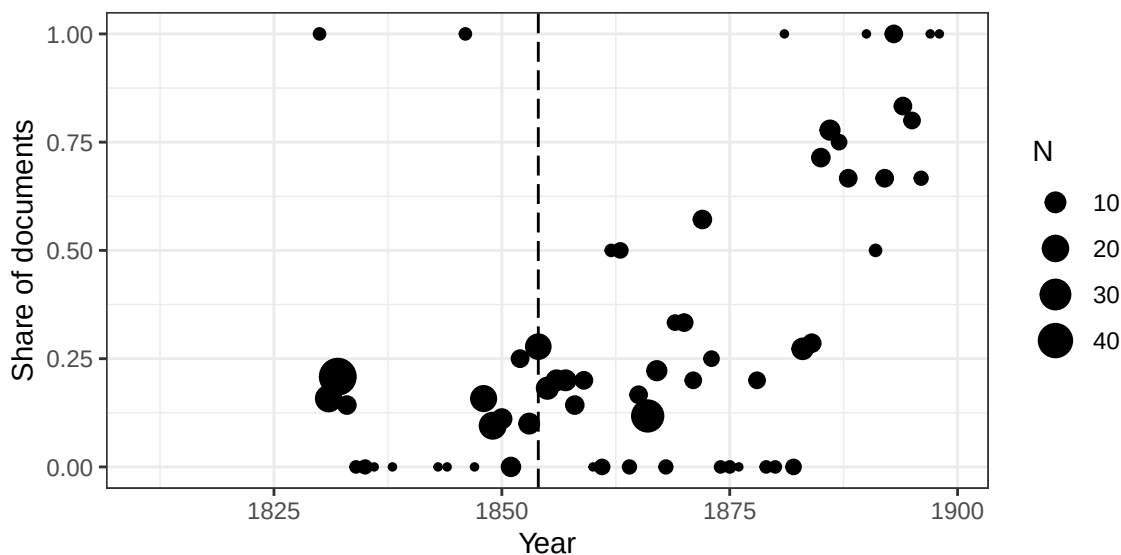


Figure A7

K Author Characteristics

Table A6

	<i>Dependent variable:</i>			
	Trade link			
	(1)	(2)	(3)	(4)
School unknown	0.059 (0.045)	0.055 (0.044)		
Scottish university			0.052 (0.052)	0.052 (0.051)
Date		-0.227 (0.186)		-0.225 (0.186)
Date ²		0.0001 (0.00005)		0.0001 (0.0001)
Observations	323	323	323	323
Adjusted R ²	0.002	0.042	0.00000	0.040

L Results for 1831-1832 Only

Table A7: Known birth dates and employment details of trade-linked authors who published in 1831-1832

Author	Age in 1831	Employment	Started Employment
Sir James Annesley	57	East India Company (active)	1799
Charles Searle	unknown	East India Company (active)	unknown
Sir Gilbert Blane	82	former physician to West Indies Fleet	retired from naval position and working in London by time of epidemic started in 1831
Frederick Corbyn	40	garrison surgeon at Fort William	
John Goss	32	East India Company (former)	unknown
George Hamilton Bell	24	Residency surgeon, Tanjore (former)	unknown
James Hutchinson	35	Surgeon to Bengal Establishment of EIC (active)	Likely sometime around 1819
Samuel Dickson	29	Assistant Surgeon, "Latel y on Madras Establishment"	unknown
Henry Marshall	56	Superintendent of medical concerns of Army in Ceylon	former
William Twining	41	East India Company (active)	1824

Table A8: Results for first wave (1831-1832) only.

	<i>Dependent variable:</i>	
	Trade link	
	(1)	(2)
Trade link	0.429 (0.140)	0.427 (0.158)
Age		−0.002 (0.004)
FRCP		0.030 (0.289)
Memberships		−0.069 (0.101)
Public audience		0.020 (0.151)
Observations	73	66
Adjusted R ²	0.095	0.042

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